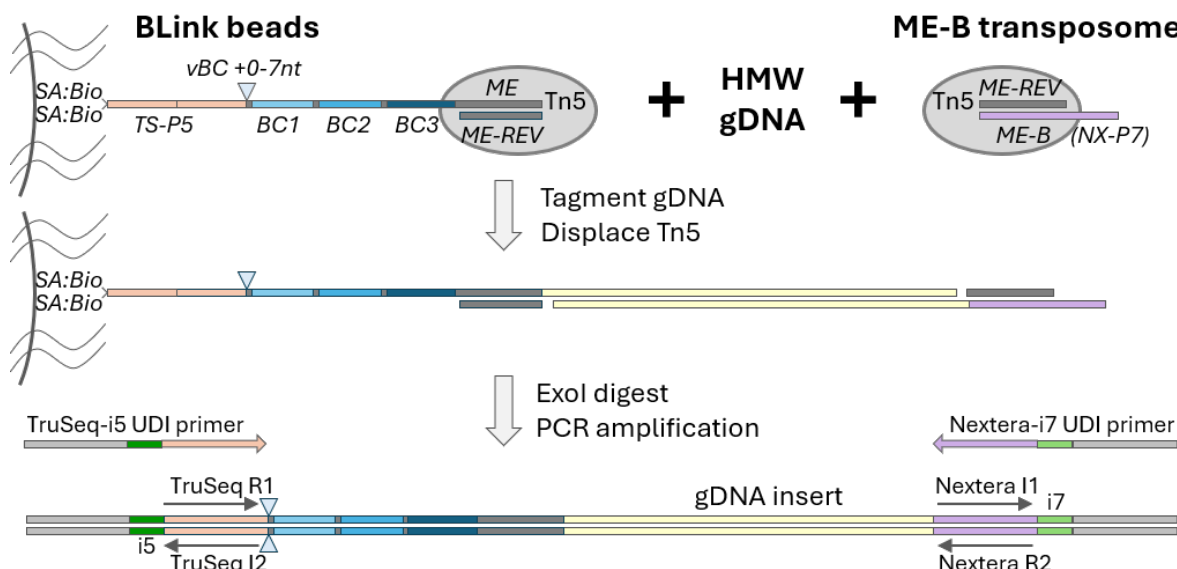


BLink-seq: Library Prep Protocol

Overview:

BLink-seq (bead-barcoded linked-read sequencing) technology generates an Illumina library of tagmented inserts that are individually barcoded based on the molecule of origin. The combinatorial bead-barcode that is added to each sub-fragment of the original HMW gDNA molecule can be used to infer phasing and linkage of sequenced fragments.

The method uses **BLink beads**: magnetic streptavidin dynabeads coupled to a 2x-biotinylated oligo containing a TruSeq-P5 PCR handle, a variable-length barcode (vBC) to offset the start of the Illumina R1 read, a combinatorial barcode (3 x 12nt BC segments), and the Tn5 ME sequence. BLink beads are loaded with purified Tn5 protein to enable on-bead tagmentation of HMW gDNA. A second Tn5 transposome in solution (ME-B) adds the Nextera-P7 PCR handle. PCR with Illumina UDI primers (TruSeq-i5 and Nextera-i7) generates the final BLink-seq library.



The final insert size distribution is determined by the frequency of on-bead tagmentation (i5 end) and solution-phase tagmentation (i7 end). Pilot experiments are recommended to optimize the insert size distribution to maximize the fraction of the library within the 450-1200bp range. Adjusting both the amount of Tn5 used to load beads (Appendix D) and especially the amount of solution-phase ME-B transposome is key to optimizing insert size. Once beads are saturated, adding more Tn5 to beads typically does not alter the insert size, but adding more ME-B transposome during library prep typically will decrease the insert size distribution. The amount of ME-B transposome required scales with gDNA input.

BLink libraries can be sequenced on Illumina instruments with standard sequencing primers and read lengths.

Materials and Equipment Required: see Appendix A

ME-B transposome: To prepare ME-B transposome in solution, see Appendix B

BLink beads: Can be stored in 3 formats:

- fully loaded with active Tn5 transposomes, ready to use: follow protocol below
- ME-REV hybridized, ready to load with Tn5: see Appendix D to prepare Tn5-loaded beads
- Beads containing barcoded oligos, but requiring final stripping and ME-REV hybridization: see Appendix C to strip/hybridize ME-REV, and then Appendix D to prepare Tn5-loaded beads

Input genomic DNA: High molecular weight (HMW) genomic DNA will generate BLink-seq libraries with more linked reads and longer N50 values. If genomic DNA is highly degraded, BLink-seq will offer little value over WGS.

Technology development: The authors of the manuscript “BLink-seq delivers population-scale haplotypes without long reads: a scalable framework for non-model genomics” contributed to developing the BLink-seq protocol.

Library Preparation Protocol

Experiment planning

Typical tagmentation reaction components and volumes:

per sample	Component	Notes
25 μ l	2x WASH+PEG buffer	final: 1x WASH, 0.1% Triton, 6.5% PEG6000
5 μ l	Tn5-loaded BLink beads	see Appendix C+D if needed
5 μ l	DNA	1.5ng total, dilute primary stocks to 0.3ng/ μ l
14 μ l	CS+DMF master mix	final: 1x CutSmart buffer, 10% DMF
1 μ l	diluted (e.g. $\sim$ 0.2 μ M) ME-B transposome*	See Appendix B if needed
50 μ l total volume in tagmentation reaction		

Volumes can be modified but the final concentrations of PEG, Triton, CutSmart buffer, and DMF should stay constant. When increasing the amount of beads per reaction, adjustments to the WASH+PEG buffer concentration can be made. For example, to use 50 μ L beads in a final 150 μ l tagmentation reaction volume, modify the plan to use a 3x formulation of WASH+PEG buffer and scale up the amount of the CS+DMF master mix to maintain the final concentration per reaction.

The ratio of DNA:beads can change to meet project needs. A higher ratio of DNA to beads will lead to more barcode 'clashing' because multiple molecules may be tagmented on the same bead and have the same barcode.

Pilot experiments: vary the amount of ME-B transposome to optimize the amount needed for a good final library size distribution. Typically the primary 2 μ M ME-B transposome stock is serially diluted in 50% glycerol storage buffer, such that 1 μ l of different dilutions are tested in the pilot phase. The optimal amount of ME-B transposome required is related to the amount of DNA, not the volume of beads; if the volume of beads is increased but DNA stays consistent, do not alter the amount of ME-B transposome per reaction.

* Once the appropriate amount of ME-B is optimized for a given batch of loaded beads + gDNA samples, ME-B transposome can be added to the CS+DMF master mix for high throughput library prep, e.g. adding 15 μ l CS+DMF+ME-B master mix per sample.

1. Prepare reagents

Recommended: use bench space and equipment that is dedicated to pre-amplification steps through step 5.

- 1x WASH+T buffer: prepare fresh each day, store at room temperature.

5mL is sufficient for $\sim$ 20 samples, scale up or prepare more as needed.

1x WASH	Stock
4.45 mL	Nuclease-free water
500 μ L	10x WASH buffer
50 μ L	10% Triton X-100
5 mL	Total volume

- 2x WASH+PEG buffer, prepare fresh each day, store at room temperature.

*prepare excess buffer, e.g. 1.2 * N samples, this buffer is viscous and may need careful handling and optimization of settings on automated liquid handlers.*

MM	per sample	Stock
	13.0 μ L	Nuclease-free water
	5.0 μ L	10x WASH buffer
	0.5 μ L	10% Triton X-100
	6.5 μ L	50% PEG8000
	25.0 μ L	Total volume

- HMW gDNA samples: dilute to target concentration in 1x WASH+T buffer (e.g. 0.3ng/ μ l in min 5 μ l).

Rotate (or shake gently < 500RPM) at room temperature to mix well. Do not vortex vigorously to avoid shearing.

While DNA samples can be diluted in advance, long-term storage of low concentration gDNA is not recommended as DNA may adsorb to polypropylene. The Triton X-100 in the 1x WASH+T buffer may help block adsorption.

2. Tagmentation reaction, followed by SDS/stop.

This protocol is designed for PCR strip tubes and reagent addition using multichannel pipettes or liquid handling robotics (e.g. Mantis). Strip tubes with individually hinged caps are recommended instead of fixed plate format to allow each strip to be mixed immediately and completely after each addition.

- Preheat a Thermomixer with 96well adapter to 55°C.
- Add 25µl WASH+PEG per sample to labeled strip tubes.
- Add 5µl Tn5-loaded BLink beads to each tube.

*Note that beads are *not* washed before use, especially if ME-B duplex oligo is added when beads are loaded with Tn5 protein (see Appendix C).*

Optional: instead of adding WASH+PEG and beads separately, prepare a master mix before aliquoting to tubes (e.g. 30µl total when using 5µl beads). This may lead to some bead loss because of the excess required to prepare in master mix format, so we recommend adding separately as indicated above if beads are limiting.

- Close the strip and flick to mix well.

The beads should be evenly dispersed before gDNA is added.

- One strip at a time, add 5µl diluted gDNA per sample. Immediately close the strip and flick/invert to mix well. Rapid and complete mixing is an important step for assay performance.

Use 1x WASH+T buffer as a no-DNA negative control – recommended in troubleshooting mode or for 1-2 wells per plate of samples.

- Incubate minimum 10 min on rotator at room temperature.

This pre-incubation step allows DNA to potentially bind to beads before magnesium is added to activate Tn5.

- Prepare CutSmart/DMF master mix at room temperature:

*prepare excess master mix, e.g. 1.1 * N samples, or more if required for liquid handling dead volume*

MM	per sample	Stock
	4 µL	Nuclease-free water
	5 µL	10x rCutSmart buffer <i>final 1x in tagmentation reaction = 10mM Mg++</i>
	5 µL	DMF <i>final 10% in tagmentation reaction</i>
	1 µL	ME-B transposome (e.g. 1:10 dilution of 2µM stock) *
	15 µL	Total

** Before the optimal amount of ME-B is determined, do not include ME-B transposome in the master mix.*

For large master mixes, a primary stock of 2µM loaded ME-B can also be used; adjust the volume of water accordingly.

- Remove the strip-tubes from the rotator. If needed, very briefly pulse-spin the strip tubes to collect the volume at the bottom of the tube but try not to pellet the beads. Flick gently if the beads are not well dispersed.

- One strip at a time, add 15µl CutSmart/DMF/ME-B Master Mix per sample. Immediately close the strip and flick to mix well.

OR:

- If ME-B is added separately (e.g. pilot phase), first add 1µl diluted ME-B transposome per sample and mix well. Then add 14µl CutSmart/DMF/ME-B Master Mix per sample and immediately mix again.

Rapid and complete mixing is an important step for assay performance. Addition of the CutSmart buffer containing magnesium will activate Tn5 transposition.

- Incubate samples 30 min, 500RPM on a Thermomixer at 55°C.
- Prepare 2% SDS (use a strip-tube reservoir, if needed; excess can be stored at room temperature).
- When the tagmentation reaction is complete, immediately add 5µl 2% SDS per sample (10% of the tagmentation reaction volume). Immediately close each strip after adding SDS and flick to mix well.
- Incubate samples 7 min, 500RPM on a Thermomixer at 55°C.
- Prepare 10% Triton X-100 (use a strip-tube reservoir, if needed; excess can be stored at room temperature).
- When the SDS incubation is complete, add 5 µl 10% Triton X-100 per sample to quench SDS (equal volume to

the amount of 2% SDS added above). Close each strip and flick to mix well.

- Pulse-spin each strip and place on a magnetic rack.

3. ExoI treatment (to remove any untagged oligos on the beads)

- Prepare ExoI master mix on ice.

*Prepare excess master mix, e.g. 1.1 * N samples, or more if required for liquid handling dead volume.*

MM	per sample	Stock
	17 μ L	Nuclease-free water
	2 μ L	10x rCutSmart buffer (NEB)
	1 μ L	Thermolabile ExoI (NEB)
	20 μ L	Total

- Prepare a multichannel reservoir with 1xWASH+T buffer.
- Remove the supernatant from the beads (on magnet) and discard. On magnet, add 100 μ L WASH+T.

Optional: if desired, beads can be 'downsampled' to generate a library for only a portion of the bead-bound tagged fragments. Downsampling will generate a library with fewer fragments and lower complexity, but can increase the percent of linked reads at lower sequencing depths. The remaining tagged beads can be stored at 4° C and processed later if desired.

- To downsample beads: remove the strip from the magnet, flick or mix to disperse the beads, and transfer the desired portion to a new strip or tube and return to magnet. The equivalent downsampling can also be done at the wash step after ExoI treatment and before PCR.

- Remove the supernatant from the beads (on magnet) and discard.

If processing many strips, either close them to prevent the beads from drying out while washing all of the strips, or add the ExoI master mix to each strip before continuing with the next strip.

- Remove the strip from the magnet and add 20 μ L ExoI master mix to each sample. Close the strip and flick to mix. Pulse-spin very briefly to ensure all of the liquid is at the bottom. *The beads should remain fully dispersed.*

- Incubate samples in a PCR machine: 10min at 37°C, then 5min at 65°C (heat kill).

If possible, set the lid temperature to 65°C.

4. PCR setup

- Thaw hybrid UDI stocks (mix of TruSeq-i5 + Nextera-i7, 5nM each) on ice. Pulse-spin when thawed.

- Prepare PCR master mix on ice:

*Prepare excess master mix, e.g. 1.1 * N samples, or more if required for liquid handling dead volume.*

MM	per sample	Stock
	12.5 μ L	2x HiFi MM (NEB #M0541)
	7.5 μ L	NF water
	20 μ L	Total

- Program a PCR machine as indicated below, including an initial pause step at 72°C. Start the PCR machine so that the block pre-heats to 72°C.

- When the ExoI incubation is complete, pulse-spin each strip and place on a magnetic rack.

Do not proceed until the supernatant is clear, the PCR master mix is prepared, UDI primers are thawed, and the PCR program is ready. The samples are stable on the magnetic rack at room temperature.

- Set a (multichannel) pipet to 100 μ L. One strip at a time on magnet, add 100 μ L WASH+T (no need to remove the ExoI reaction first – just add WASH+T to the tube). If the beads come off the magnet, pause until the supernatant is clear. With the same tips, remove the entire supernatant (~120 μ L).

If processing many strips, close them to prevent the beads from drying out while washing all of the strips.

Optional bead downsampling:

To downsample beads: remove the strip from the magnet after adding WASH+T buffer, and mix to fully disperse the beads. Transfer the desired volume to a new strip and place on a magnetic rack. When the supernatant is clear, remove the supernatant. *The remaining tagged beads can be stored at 4° C and processed later if desired.*

- Move the strips containing the bead pellets to a chiller block or a rack on ice.
The PCR mix is not a hot-start formulation, because the tagmented ends need to be extended before the first denaturation step. Keep all PCR reagents and the samples on ice while setting up the PCR reaction, and transfer directly to a pre-heated PCR block to minimize polymerase activity below 72°C.

- On a chiller block or on ice, add 20 µl 2x HiFi Master Mix per sample.
- On a chiller block or on ice, add 5 µl hybrid UDI stock (TruSeq-i5 + Nextera-i7, 5nM each) to each sample.
Important: record which UDI wells are used for each sample.

- PCR: Initial pause: 72°C *preheat and hold at temperature*
Initial incubation: 72°C/5min *extend tagmentation gaps*
Initial denaturation: 98°C/30sec
Cycling: [98°C/10sec – 65°C/30sec – 72°C/1min] x 12 cycles*
Final extension: 72°C/5min

** PCR cycle number may need to be adjusted based on beads and DNA input.*

Recommended: put away all pre-PCR reagents before moving to a different bench for post-PCR cleanup.

5. Post-PCR cleanup and quality checks

- Prepare fresh 80% ethanol (500µl per sample).
- Place PCR reactions on post-PCR magnetic rack.
- Set up new strips with 22.5 µl DNA cleanup beads (0.9x).
- Transfer PCR reaction to new strips containing DNA cleanup beads.
- Vortex to mix well, incubate at least 5min at room temperature.
- Pulse-spin and transfer strip to magnet.
- On magnet, wash 2x with 200 µl fresh 80% ethanol.
For each wash, pipette up and down (on magnet) to rinse the beads, then leave the ethanol wash on the beads for minimum 30sec.
- After 2nd wash: pulse spin and remove residual wash.
- Allow beads to dry briefly, only until droplets are gone (beads should still be dark and glossy).
- Remove from magnetic rack and resuspend beads in 23µl 0.1x TE.
- Incubate at least 5 min at room temperature, transfer to magnetic rack until clear.
- Transfer supernatant to a clean, labelled strip tube or plate. This is the final BLink-seq library.

6. Optional: quantify and check the size distribution for all or a subset of samples.

- Prepare tubes for Qubit assay: add 198 µl 1x HS dsDNA Qubit reagent, hold in the dark at room temperature until equilibrated.
- Transfer 2µl final library stock to Qubit tubes. Vortex, pulse-spin, and incubate 2 min at room temperature.
- Read and record Qubit measurements.
- Prepare aliquots for size QC, dilute if needed to be in the target concentration range.

Final library size can be estimated by gel electrophoresis or using an instrument such as the Fragment Analyzer (Agilent) or equivalent. A successful library will typically have a size range from ~300-1500bp, with 40% or more in the target size range of 450-1200bp. To adjust the size range in the next experiment, alter the amount of ME-B transposome used (more ME-B transposome = smaller insert sizes and *vice versa*). Libraries with less material in the target size range can still be sequenced, but the amount of linked reads may be lower and/or the insert size may be smaller than ideal (<450bp).

7. Pooling and sequencing

- Pool final libraries, based on QC metrics (Qubit concentration and size distribution) when available. If the final pool will be size-selected, plan the pool based on molarity estimates for the target size range.
- Recommended: size-select the final pool (450-1200bp) on a Pippin instrument (Sage Science) in order to optimize

insert sizes prior to sequencing.

- Sequence on an Illumina sequencer, recommended 2x150 PE reads with dual index reads. Index read lengths should match the UDI primers used.

Analysis with Harpy: see <https://pdimens.github.io/harpy/>.

Harpy includes several modules for preprocessing reads, aligning reads, calling variants (SNPs and SVs), and more.

Appendix A: Materials and Equipment Required

Reagents and Supplies:

Tn5-loaded BLINK beads	<i>see Appendix C and D if needed</i>
Unloaded Tn5 protein	Diagenode #C01070010 <i>required to load ME-B transposome and beads (see Appendix B and D)</i>
10x rCutSmart buffer	New England Biolabs #B6004S <i>Other Tn5 tagmentation buffers (final 10mM Mg⁺⁺) are likely to work as well</i>
Thermolabile ExoI	New England Biolabs #M0568S/L
2x High Fidelity PCR Master Mix	New England Biolabs #M0541S/L
Tris pH8	<i>available from many suppliers</i>
NaCl	<i>available from many suppliers</i>
EDTA	<i>available from many suppliers</i>
Glycerol	<i>available from many suppliers</i>
SDS	<i>available from many suppliers</i>
Triton X-100	<i>available from many suppliers</i>
PEG8000	50% in water: Hampton Research #HR2-535, or resuspend from powder
Dimethylformamide	<i>available from many suppliers</i>
Nuclease-free water	<i>available from many suppliers</i>
Magnetic DNA cleanup beads	e.g. SparQ beads (QuantaBio / VWR) or equivalent, including  speedbeads
100% Ethanol	<i>available from many suppliers</i>
Qubit 1X dsDNA HS Assay Kit	Invitrogen #Q33231

Custom primers, shown 5'→3' orientation:

Tagmentation oligos

Needed for all library preps, order large synthesis scale

Resuspend at 500nM stock concentration in 0.1x TE (1mM Tris pH 8, 0.1mM EDTA)

ME-B GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAG

ME-REV-X /5Phos/CTGTCTCTTATACACATCT/3InvdT/

recommended but not required: add 3'Inverted-dT to block 3' extension of the ME-REV oligo

Illumina UDI PCR primers

Order in plate format at normalized concentration, e.g. 50 µM each

TruSeq-i5 PCR primer AATGATACGCGACACCAGAGATCTACAC-NNNNNNNN-ACACTCTTTCCCTACACGACGCTCTTCCGATCT

Nextera-i7 PCR primer CAAGCAGAAGACGGCATACGAGAT-NNNNNNNN-GTCTCGTGGGCTCGGAGATGTGTATAAGAGAC

NNNNNNNN represents the sample barcode, 8-10bp barcodes recommended.

Ready to use: prepare a hybrid UDI stock of each primer pair (TruSeq-i5 + Nextera-i7, 5nM each)

Equipment and supplies:

Pipettors (including multichannels), vortex (tube and plate), mini-centrifuge (tube, strip tube, and plate), Thermomixer with tube and 96-well plate adapters, tube and plate rotator, PCR machine, magnetic racks (tube and plate), Qubit, FragmentAnalyzer/Tapestation/GXII Touch, BluePippin/PippinHT.

Liquid handling robots can aid with large batches. A Mantis (Formulatrix) is particularly useful for non-contact addition of master mixes.

Tubes (1.5mL, 5mL, 200ul PCR strip tubes, 15mL and 50mL falcon tubes), plates, filtered pipet tips, reagent reservoirs

Stock buffers:

10x WASH buffer 100mM Tris pH 8, 300mM NaCl

Appendix B: Preparation of ME-B transposome in solution

Additional materials (see also Appendix A):

HEPES pH 7.3 ThermoFisher # J16924 *or equivalent*

Buffers:

10x Annealing buffer 100mM Tris pH 8, 500mM NaCl

50% glycerol storage buffer 50 mM HEPES, 0.1M NaCl, 0.1mM EDTA, 0.1% Triton X-100, 50% glycerol

Protocol:

If you have already prepared ME-B/ME-REV duplex oligo stock, start from step 3. Otherwise, start from step 1.

1. Prepare an equimolar mixture of ME-B and ME-REV oligos in 1x Annealing buffer in a PCR tube.

For example, to prepare 100 μ l of 50 μ M ME-B/ME-REV duplex oligos, combine:

70 μ l nuclease-free water

10 μ l 10x Annealing buffer

10 μ l 500 μ M ME-B oligo

10 μ l 500 μ M ME-REV oligo

2. Heat and slowly cool the oligos to promote duplex formation: 65°C/5min – slow cool 1°C/minute to 22°C.

The ME-B/ME-REV duplex oligo stock can be stored at -20°C. If dilution is required, dilute in 1x Annealing buffer.

3. Prepare the Tn5 loading reaction as an equimolar mixture of ME-B/ME-REV duplex oligo and unloaded Tn5 protein in 50% glycerol storage buffer.

For example, to prepare 50 μ l of 2 μ M ME-B/ME-REV loaded Tn5, combine:

47 μ l 50% glycerol storage buffer

2 μ l 50 μ M ME-B/ME-REV duplex oligo stock

1 μ l unloaded Tn5 (Diagenode ~100 μ M stock)

4. Rotate the loading reaction at room temperature for 30 minutes. Optionally, continue to rotate overnight at 4°C. Store the loaded stock at -20°C.

Appendix C: Strip and anneal ME-REV on BLink beads

Additional materials (see also Appendix A):

NaOH	<i>available from many suppliers</i>
D-Biotin	<i>Invitrogen #B1595 or equivalent</i>
DMSO (dimethylsulfoxide)	<i>available from many suppliers</i>

Buffers:

NaOH stripping buffer	0.15 M NaOH, 0.1% Triton
200mM Biotin	Dissolve 4.9mg D-Biotin per 100µl DMSO
WASH+T buffer	<i>See main protocol</i>

Protocol: Strip duplex oligos from beads and hybridize the ME-REV oligo

This section starts with BLink beads that have completed the split-pool ligation steps but still have the full barcode-duplex oligos. The reverse-strand oligos do not get ligated to each other because there is no 5' phosphate, and are not attached to the beads other than by the duplex hybridization. These oligos are removed with 2 rounds of NaOH denaturation and wash steps. Then the ME-REV oligo is hybridized to the bead stock prior to the final step of loading Tn5 protein (Appendix D).

In the standard stripping protocol below, 415µl of the ligated bead stock is stripped with 1mL 0.15M NaOH. This can be scaled down if desired, or scaled up by processing multiple tubes of 415µl of the ligated beads in parallel.

These steps require 2 Thermomixers with 1.5mL blocks, one set to 55°C and a second set to 40°C. If only 1 Thermomixer is available, remove the block from the base immediately after the final incubation at 55°C to promote rapid cooling to 40°C.

1. Prepare buffers: volumes indicated are sufficient to strip 1 tube of 415µl of ligated bead stock, and can be scaled as needed for a smaller volume of beads or for multiple tubes stripped in parallel.
 - Prepare 2.2 ml NaOH stripping buffer. Hold at room temperature.
 - Prepare 6 ml WASH+T buffer.
 - Transfer 1.2mL WASH+T buffer to a new tube and pre-warm in a Thermomixer at 55°C. *Use the heated lid if available.*
 - Transfer 1.1 mL WASH+T buffer to another tube and add 11µl 100 µM ME-REV oligo. Hold on ice. *The final concentration is 1 µM ME-REV oligo.*
 - Save the remaining WASH+T buffer at room temperature.
2. Mix the ligated bead stock by inversion or rotation until fully dispersed in suspension.
3. Transfer 415µl to new 1mL tubes and place on magnet.

The timing of the stripping steps should not be extended: if processing more than 4 tubes in parallel, consider processing in smaller batches to avoid excessive incubation in NaOH.
4. Remove all WASH+T buffer from the beads and remove from magnet.
5. Add 1 mL NaOH stripping buffer. Immediately close the tube and mix by inverting several times until no bead pellet is visible. Incubate the tube on a rotator for max 1 min at room temperature, then pulse spin and place on magnetic rack (the beads should be in contact with NaOH for at most 2 minutes).
6. On magnet, as soon as the beads are pelleted, remove all NaOH stripping buffer (save the waste for appropriate disposal).
7. Remove the tube from the magnetic rack and immediately add 1.2 mL **warm** WASH+T buffer. Mix by inverting until no bead pellet is visible.
8. Rotate beads in WASH+T buffer for 1 min at room temperature; pulse-spin and return to magnet.

9. While beads are washing:

- Refill the pre-warmed WASH+T tube to 1.2mL and pre-warm on the Thermomixer at 55°C.
 - Pre-warm the tube of WASH+T buffer containing 1 μ M ME-REV to 55°C in the Thermomixer.
10. Repeat NaOH stripping steps:
- On magnet, remove WASH+T buffer from the beads
 - Add 1mL NaOH stripping buffer, mix, rotate max 1min at room temperature, pulse spin, and return to magnet.
 - As soon as beads have pelleted, remove NaOH stripping buffer (save the waste for appropriate disposal)
 - Add 1.2mL **warm** WASH+T, rotate 1 min at room temperature, pulse and return to magnet.
11. When beads have pelleted, remove all WASH+T buffer and add 1 mL **warm** WASH+T buffer with 1 μ M ME-REV.
12. Rotate the tube for 5 min at room temperature.
13. Transfer the tube to a Thermomixer set to 40°C for 5 min at 800 RPM.
14. Return to the rotator at room temperature and continue to mix for another 15 min.
15. Prepare WASH+T+Biotin buffer: in a new tube, combine 1.4mL WASH+T buffer and 3.5uL 200mM Biotin. Vortex to mix; hold at room temperature.
16. After the 15 min incubation, pulse-spin the tube containing the hybridized beads and place on a magnet.
17. When beads have pelleted, remove the WASH+T+ME-REV buffer.
18. Add fresh 1.2 mL Wash+T+Biotin buffer and rotate for another 20 min at room temperature.

The tube now contains beads ready for Tn5 loading (see Appendix D).

Store the stripped beads hybridized with ME-REV in WASH+T+Biotin buffer at 4°C.

Appendix D: Load BLink beads with Tn5

Additional materials (see also Appendix A):

HEPES pH 7.3 ThermoFisher # J16924 or equivalent

Buffers:

WASH+T buffer *See main protocol*
50% glycerol storage buffer *For Tn5 protein dilution: see Appendix B*
30% glycerol storage buffer 50 mM HEPES, 0.2M NaCl, 0.2mM EDTA, 0.2% Triton X-100,
30% glycerol

Part I Small scale Tn5 loading to find the optimal Tn5 concentration

Goal: Titrate amount of Tn5 protein to achieve optimal tagmentation of gDNA to approximately 450-1200bp library size.

This section starts with BLink beads that have been hybridized with ME-REV and are ready to load Tn5. The protocol describes loading 4 different Tn5 concentrations, but can be scaled to as many test loads as desired. Typically, 25µl of the ME-REV hybridized bead stock is loaded in a final volume of 50 µl.

ME-B/ME-REV duplexed oligos (ME-B duplex) can also be added during the bead load. While this ME-B is not sufficient to generate the appropriate insert sizes for Illumina sequencing, it may help to make tagmentation more efficient on the bead surface. ME-B transposome is supplemented during library prep as well. See Appendix B to prepare 50 µM ME-B/ME-REV duplexed oligo stock; make additional dilutions in 1x Annealing buffer after duplex formation.

Diagenode unloaded Tn5 protein is very concentrated (approx. 100µM), and can be diluted to prepare a working stock in 30% or 50% glycerol storage buffer. If preparing excess diluted unloaded Tn5 protein stock, dilute in 50% glycerol storage buffer and store the remainder at -20 °C. The typical loading amount is ~8nM Diagenode unloaded Tn5 protein, or 4µl of a 1:1000 dilution of the primary unloaded Tn5 stock per 50µl bead load.

Example for 4 test loads:

Test load #				Bead loading reagents
1	2	3	4	
46	45	44	43	µl 30% glycerol storage buffer
1	1	1	1	µl 0.2µM /ME-REV duplex oligos (optional)
3	4	5	6	µl unloaded Tn5 protein (1:1000 dilution)

- Mix the stock of BLink beads hybridized with ME-REV on a rotator until fully dispersed.
It can help to lift the beads out of the bottom of the tube on a magnetic rack before placing on the rotator.
- Transfer 100µl to a new 1.5 ml tube (*sufficient for 4 tests using 25µl orig beads/test*) on a magnetic stand.
- When the beads have pelleted on the magnet and the WASH+T buffer is clear, remove the WASH+T buffer.
- Remove the tube from the magnetic rack and add 105µl of WASH+T buffer. Mix well.
- Aliquot 25µl into 4 wells of a new PCR strip and place on a magnetic stand.
- When the beads have pelleted on the magnet and the WASH+T buffer is clear, remove the WASH+T buffer.
- Remove the strip from the magnetic rack and add 30% glycerol storage buffer (volume based on the test loading plan). Close the strip and flick to disperse the beads.
- Optional: add ME-B duplex oligos to each tube, final concentration 4nM. Close the strip and flick to mix.
- Add unloaded Tn5 protein (diluted) to bring volume to 50 µl. Immediately close the strip and flick to mix well.
- Rotate at least 10 minutes at room temperature, then overnight (or longer) at ~4°C. Beads may continue to load over several days.

Store loaded beads at 4°C.

11. Follow the 'BLink-seq: library prep protocol' with a control gDNA sample to test Tn5 loading, using 5µl of the Tn5-test loaded beads per library. It is also important to titrate the ME-B transposome (in solution) to optimize yield and fragment size distribution, e.g. prepare libraries with 2 or more ME-B dilutions per bead test load. Typically, bead performance will plateau and adding more Tn5 will not alter library yield or size distribution above the optimal loading concentration.

Part 2: Batch loading of Tn5

After determining the optimal amount of Tn5 to use per 50µl bead load, the loading protocol can be scaled up as desired. Maintain the same Tn5:bead ratio, scaling the volume up to max 500µl loaded beads per 1.5mL tube. The final volume of loaded beads will be double the starting volume of hybridized, unloaded beads.

For example:

1. Mix the stock of BLink beads hybridized with ME-REV on a rotator until fully dispersed.
It can help to lift the beads out of the bottom of the tube on a magnetic rack before placing on the rotator.
2. Transfer the desired volume of beads to one or more 1.5mL tube (max 250µl hybridized bead stock per tube, to load in max 500µl 30% glycerol storage buffer).
3. Place on a magnetic rack. When the beads have pelleted on the magnet and the WASH+T buffer is clear, remove the WASH+T buffer.
4. Remove the tubes from the magnetic rack and add 500µl fresh WASH+T per tube. Mix well to disperse the beads.
5. Pulse spin and place the tubes back on the magnet.
6. Prepare a loading master mix containing 30% glycerol storage buffer, ME-B duplex (optional), and unloaded Tn5 protein, with total volume sufficient for the batch load.
7. Remove the WASH+T from the beads and add up to 500µl loading master mix. Immediately mix to disperse the beads.
8. Place the tube(s) on rotator at room temperature for minimum 10 minutes, then rotate overnight or longer at 4°C.
The loading performance appears to improve after incubating at 4°C for several days.
9. Pulse spin down the tubes. If several tubes are loaded per batch, they can be mixed together to ensure the batch is homogeneous. **Store Tn-5 loaded BLink beads at 4°C.**
10. Recommended: run final library prep tests to confirm the batch is performing as expected. The amount of ME-B transposome required during library prep may need to be optimized for each batch.